



Routine metabolism of larval green sturgeon (*Acipenser medirostris* Ayres)

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Abstract

Routine metabolic rates of green sturgeon (*Acipenser medirostris*) from hatching to 31 days post hatch (dph) were determined under normoxic conditions. During the endogenous feeding stage that comprised period from hatching to 15 dph, the oxygen consumption rate (MO_2 , $\mu\text{g O}_2 \text{ larva}^{-1} \text{ h}^{-1}$) increased 5-fold before yolk reserves became exhausted and MO_2 rates steady. The allometric relationships between MO_2 and body mass had mass exponents greater than 1.0 ($b = 1.64 \pm 0.21$) and equal to 1.0 ($b = 1.04 \pm 0.07$) during the endogenous and exogenous feeding phases, respectively. The magnitude and changes of MO_2 rates in green sturgeon larvae reflected their early ontogeny, especially during the endogenous feeding phase when the increase in metabolic rates was associated with organogenesis, acquisition of organ functions, and the conversion of yolk sac into new tissues.

Introduction

The anadromous green sturgeon is considered a rare or vulnerable species in the United States and Canada (Birnstien 1993) and an endangered species in Russia (Artyukhin and Andronov 1990). Published information on this species is very limited, and we are currently investigating life history and environmental physiology of green sturgeon, using artificial spawning techniques and cultured fish (Van Eenennaam et al. 2001).

The only known spawning populations of green sturgeon in North America are in the Klamath, Rogue and Sacramento Rivers, all of which are affected by water projects (Moyle et al. 1994). Changes in flow of rivers controlled by dams, especially during the reproductive season of green sturgeon (late spring, Van Eenennaam et al. 2001), may be detrimental for reproduction, in regards to both water quality (temperature and dissolved oxygen) and availability of food resources for green sturgeon larvae. Khakimullin (1987) reported that Siberian sturgeon (*A. baerii*) larvae during their endogenous feeding phase could not avoid

hypoxic areas, which directly affected their survival. There is significant information in Russian literature (reviews of Klyashtorin 1982; Gershanovich et al. 1987) on the respiratory metabolism of juvenile sturgeon and paddlefish, but only limited information exists on the oxygen consumption of larval stages (Khakimullin 1984; Dabrowski and Kaushik 1986).

The objective of this study was to measure the routine respiratory metabolic rate of green sturgeon larvae from hatching to onset of metamorphosis under normoxic conditions. Knowledge of oxygen consumption patterns during the larval development (including endogenous and exogenous feeding phases) will provide basic information for further studies on green sturgeon larval physiology and for the projects aiming to protect reproductive habitat of green sturgeon.

Materials and methods

Fish supply and maintenance

Larvae were obtained by induced spawning of one female and male green sturgeon that were caught

by gillnets and held in cages in the Klamath River (Weitchpec, California). Fish were induced to ovulate and spermiate by injections of gonadotropin releasing hormone as previously described (Van Eenennaam et al. 2001). Ovulated eggs were fertilized *in vitro* (fertilization rate 78% at early cleavage) and, after bathing in silt to eliminate adhesiveness, were transported to the Center of Aquatic Biology and Aquaculture (CABA, UC Davis) and incubated in MacDonald jars at 13–15 °C for seven days before hatching. Five hundred hatched larvae were transferred into a circular fiberglass tanks (120 l) supplied with well water in a temperature-controlled recirculation system. Fish were held at a constant temperature of 16 °C and exposed to 12L:12D artificial photoperiod. Food was presented at 12 days post hatch (dph) and larvae were fed with a semi-moist commercial diet (Silver Cup, Nelson & Sons, Inc., Murray, UT) mixed with live chopped *Tubifex* worms. Once first feeding was detected between 14 and 16 dph, larvae were fed *ad libitum* and the excess of uneaten feed and feces were removed daily. Prior to the measurement of routine metabolic rates, larvae were fasted for 16 h in a separate tank in order to avoid a postprandial increase in oxygen consumption (Dabrowski et al. 1987).

Respirometry

Measurements of oxygen partial pressure (PO₂) were made using an O₂ analyzer (PHM71/MK2 Radiometer, Copenhagen). Static-type respirometers (glass syringes, approximate volume: 60 ml; Micro-Mate Interchangeable, Popper & Sons, Inc. USA) were used to measure routine metabolic rates of fish (Fry 1971). The advantages of using this type of respirometer with fish larvae are described in Marty et al. (1990). Water samples were injected directly from the respirometer into a thermostatted chamber that contained the calibrated O₂ electrode. Twelve assays were carried out each day from 09:00 to 13:00 h in order to measure the oxygen consumption rates of larvae. For each experiment, six respirometers were situated in an aerated, temperature-controlled (16 ± 0.1 °C) fiberglass water bath similar to general rearing tanks. Two respirometers were used as a blank (no fish) to correct for potential microbial oxygen consumption. In all experiments, microbial oxygen consumption was negligible.

Prior to each experiment, fish were quickly transferred from the tank to the respirometers with a dip net in order to minimize handling stress, and were

allowed 1-h acclimation before oxygen consumption measurements. Preliminary assays indicated that this time was fully adequate for the complete acclimation of fish (metabolic plateau) to the respirometer. After acclimation, the initial PO₂ (> 80% air saturation) was measured by injecting 5 ml of water (remaining water volume in the respirometer was 50 ml) directly into the oxygen electrode chamber. The static-type respirometers were left undisturbed on the bottom of the experimental water bath for one hour. At the end of each experiment, water in the respirometer was gently mixed by slowly inverting the syringe, and 5 ml were injected into the oxygen electrode chamber. Sample PO₂s (mm Hg) were converted to oxygen concentrations (mg O₂/l) by using the nomogram of Green and Carrit (1967). Oxygen concentration at the end of each experiment was > 60% air saturation and never reached the limits of optimal range (52–58% air saturation) known in larvae of other species, such as Siberian sturgeon and stellate sturgeon, *A. stellatus* (Lozinov 1956 quoted by Khakimullin 1987).

Oxygen consumption (including mass-specific) rates were calculated from the rate of decrease in water of PO₂ in the respirometer, respirometer volume, time between initial and final measurements, number and total mass of larvae in each respirometer (Barrionuevo and Burggren 1999). The number of larvae placed in each respirometer decreased with increasing age and size to avoid confining-stress effects on oxygen consumption rates (five larvae from hatching to 7 dph, three from 8 to 23 dph, two from 24 to 28 dph, and one from 29 to 31 dph). After each series of metabolic measurements, fish were anaesthetized with tricaine methanesulphonate (MS 222) and wet body weight (BW) measured to the nearest 0.01 mg using an electronic balance. For comparing green sturgeon routine oxygen consumption rates (MO₂) with those already reported in other species, data for endogenous and exogenous feeding phases were expressed as allometric functions, $MO_2 (\mu g O_2 larva^{-1} h^{-1}) = a BW^b$, where BW is a wet weight of larva (Rombough 1988a). The relationship between the age and the oxygen consumption rate was determined by fitting linear regression using SigmaStat 2.0 (Jandel Scientific). The accepted significance level was $P < 0.05$. Data in text is mean ± SE.

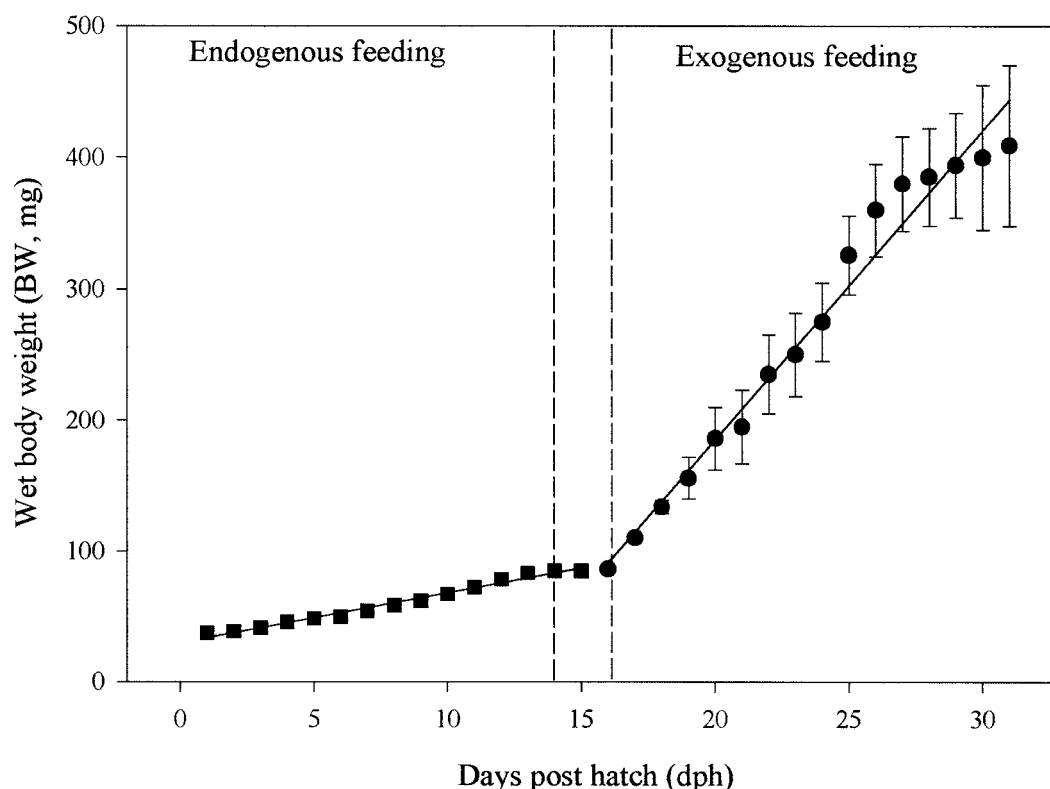


Figure 1. Wet body weight (mean $\pm$ SD, $n = 10$) of green sturgeon larvae during 31 days post hatch. (■) Yolk-sac larvae; (●) *ad libitum*-fed larvae. Dashed line represents the period of transition to exogenous feeding. The SD bars during the endogenous feeding phase are not visible due to their small value.

Results

Growth in wet weight of green sturgeon larvae could be divided in two different phases according to linear regression results (Figure 1): a lecithotrophic (yolk feeding) phase between hatching and onset of exogenous feeding at 15 dph (BW mg = $30.0 + 3.8$ dph, $r^2 = 0.98$) and an exogenous feeding phase between 16 and 31 dph (BW mg = $-285.0 + 23.6$ dph, $r^2 = 0.97$).

During the endogenous feeding phase, the MO_2 ($\mu\text{g O}_2 \text{ larva}^{-1} \text{ h}^{-1}$) increased 5-fold, from $8.7 \mu\text{g O}_2 \text{ larva}^{-1} \text{ h}^{-1}$ ($0.233 \mu\text{g O}_2 \text{ mg}^{-1} \text{ h}^{-1}$) at hatching to $44.9 \mu\text{g O}_2 \text{ larva}^{-1} \text{ h}^{-1}$ ($0.529 \mu\text{g O}_2 \text{ mg}^{-1} \text{ h}^{-1}$) at 14 dph (Figure 2). The large increase in MO_2 occurred from hatching to 10 dph ($\text{MO}_2 \mu\text{g O}_2 \text{ larva}^{-1} \text{ h}^{-1} = 3.4 + 4.34$ dph; $r^2 = 0.99$), with a plateau at the end of yolk resorption (11–15 dph). During the exogenous feeding phase, the MO_2 increased from $48.3 \mu\text{g O}_2 \text{ larva}^{-1} \text{ h}^{-1}$ ($0.567 \mu\text{g O}_2 \text{ mg}^{-1} \text{ h}^{-1}$) at 15 dph to $155.3 \mu\text{g O}_2 \text{ larva}^{-1} \text{ h}^{-1}$ ($0.451 \mu\text{g O}_2 \text{ mg}^{-1} \text{ h}^{-1}$) at 28 dph, and reached

$322.7 \mu\text{g O}_2 \text{ larva}^{-1} \text{ h}^{-1}$ ($0.723 \mu\text{g O}_2 \text{ mg}^{-1} \text{ h}^{-1}$) at 31 dph (Figure 2). The allometric relationships between the individual MO_2 and body mass revealed increased rate of oxygen consumption ($b = 1.64 \pm 0.21$) during the endogenous feeding phase, and close to the unity regression coefficient ($b = 1.04 \pm 0.07$) during the exogenous feeding larvae (Figure 3).

Discussion

Green sturgeon produce large yolk eggs, investing greater amount of energy for nourishment of the embryo, compared to other acipenserids (Van Eenennaam et al. 2001; Deng et al. 2002). At hatching, most functional systems of green sturgeon larvae are incomplete; consequently, significant organogenesis and acquisition of organ functions occur during the larval development (Deng et al. 2002). Similar to modern teleosts, sturgeon embryo utilizes ca. 20% of yolk proteins before hatching, conserving yolk nutrients for continuing organogenesis and as a source of energy

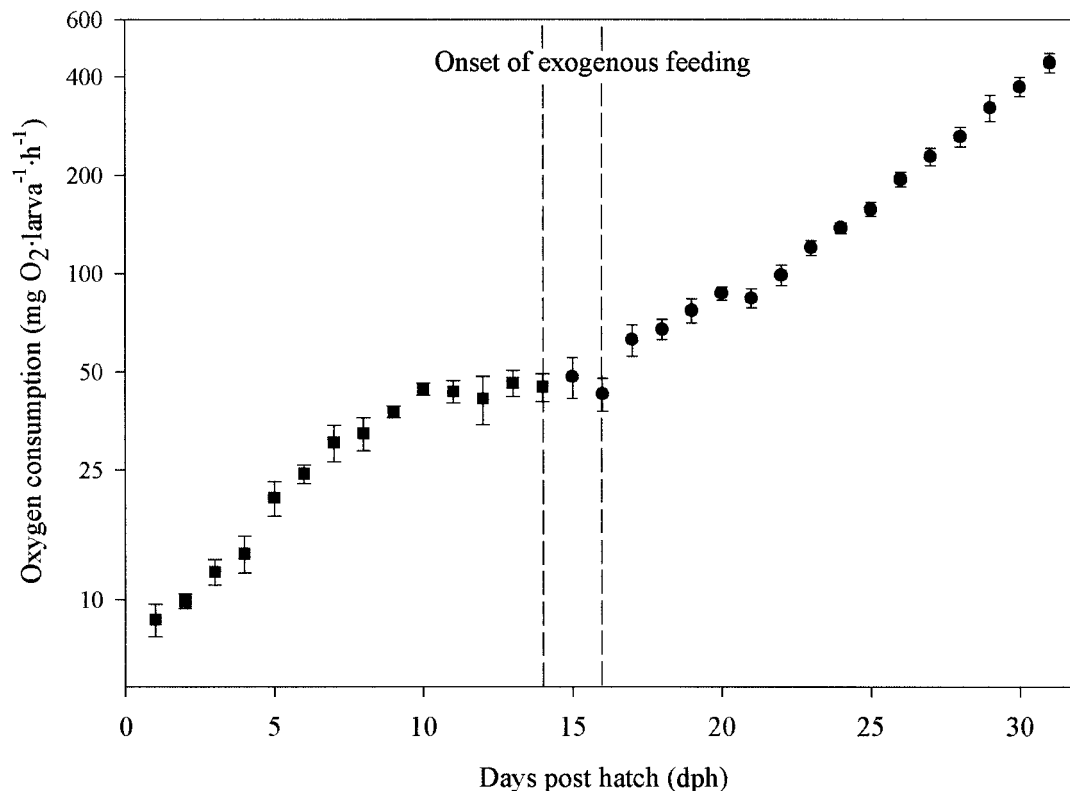


Figure 2. Larval oxygen consumption rates of green sturgeon during the endogenous (■) and exogenous (●) feeding phases. Dashed line represents the period of transition to exogenous feeding. Symbols are mean $\pm$ SE from 10 respirometry measurements.

during the larval stage (Wang et al. 1987). Organogenesis and replacement of embryonic structures with the functional adult organs in lecithotrophic phase of sturgeon larval development (Schmalhausen 1991; Gisbert 1999) may be reflected in larval energetics and metabolism. During the endogenous feeding phase, the MO_2 of green sturgeon larvae increased 5-fold before yolk sac reserves became limiting and metabolic rates steady. The large increase in MO_2 rate from 5 to 10 dph coincided with the disappearance of embryonic vascular system in the posterior region of endodermal yolk sac (involved in cutaneous respiration and nutrient transport), and with the onset of branchial respiration (rhythmic movements of mandible and operculum). Similar developmental changes in the MO_2 during endogenous feeding were reported in Siberian sturgeon (Dabrowski and Kaushik 1986) and salmonids (Lukina 1983; Rombough 1988b; Rombough and Ure 1991), while in common carp the transition from cutaneous to branchial respiration resulted in a decrease in MO_2 rates (Kamler 1972). Similar to Siberian sturgeon (Dabrowski and Kaushik 1986),

the increase in metabolic rate of green sturgeon during the endogenous feeding stage is likely associated with the conversion of yolk proteins into new larval tissue and acquisition of definitive organ functions (e.g., gill respiration, digestive system development).

In feeding larvae, the MO_2 increased, after a short-term plateau, to the end of rearing. The almost two-fold increase in MO_2 observed between 29 and 31 dph could be associated with metamorphic events (Kamler 1992) or with the confining effect (fish size greatly increased at this age) of the respirometer (Cech et al. 1985). In addition, we used a single individual in the respirometer during 29–31 dph, and it has been shown that MO_2 is lower when measured in the small groups (vs. individual fish) of larval and juvenile sturgeon (Khakimullin 1988) and paddlefish, *Polyodon spatula* (Gershanovich et al. 1987).

When expressing the MO_2 as an allometric function ($MO_2 = aBW^b$) for the entire experimental period, the mass exponent b was close to the unity ($b = 1.14 \pm 0.1$), but developmental differences between endogenous and exogenous feeding phases were ob-

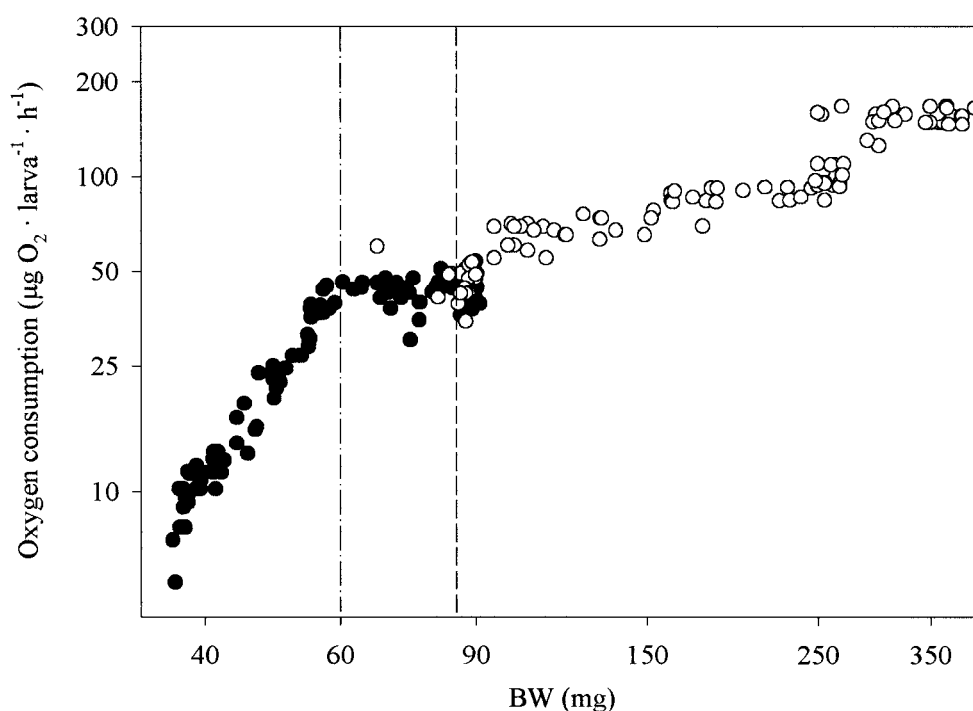


Figure 3. Relationship between oxygen consumption rates and wet body weight during the endogenous (●) ($\text{MO}_2 = 0.04 (0.03)\text{BW}^{1.64(0.21)}$; $r^2 = 0.87$; $P < 0.001$; $n = 140$) and exogenous (○) ($\text{MO}_2 = 0.40 (0.15)\text{BW}^{1.04(0.07)}$; $r^2 = 0.97$; $P < 0.001$; $n = 140$) feeding phases of green sturgeon larvae (see text for details). The dashed-dotted line shows the average weight (60.0 ± 2.1 mg) of the complete transition to branchial respiration (9–10 dph); dashed line represents the average weight (85.6 ± 1.7 mg) at the onset of exogenous feeding (14–15 dph).

scured (Rombough 1988a). In the separate equations, mass exponent was greater than 1.0 ($b = 1.64 \pm 0.2$) in the endogenous feeding phase and did not differ from 1.0 ($b = 1.04 \pm 0.1$) in the exogenous feeding phase. Khakimullin (1984) reported similar difference between two phases (mass exponents 1.31 and 0.85 for endogenous and exogenous feeding phases, respectively) in larval development of Siberian sturgeon. Gershanovich et al. (1987) reported mass exponents ranging 0.79–0.98 for feeding larvae and juveniles of several species of sturgeon and paddlefish, which is similar to our results for exogenous feeding phase. Rombough (1988a) reviewed the allometric relationships in more than 25 teleosts, and concluded that high mass exponents could be a reflection of favorable, for the respiratory exchange, surface to volume ratios in the early life stages. However, metabolic mass exponents were seldom the same for different species, because they are affected by the experimental design, fish activity and environmental factors such as temperature.

In conclusion, the magnitude and changes of MO_2 rates in green sturgeon larvae reflected their early

ontogeny, especially during the endogenous feeding phase when the increase in metabolic rates seemed to be associated with morphogenesis and the conversion of endogenous nutrients into new tissues. During the exogenous feeding phase, routine metabolic rates increased in a direct proportion to the increase in larval weight as indicated by the metabolic mass exponent equal to unity.

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